MECWIN®

EMPOWERING FUTURE

HIGH-EFFICIENCY HIGH-VOLTAGE FLAME-PROOF THREE-PHASE ASYNCHRONOUS MOTOR



www.mecwinindia.com

COMPANY OVERVIEW

About Us

Mecwin Group is a Bengaluru-based engineering and manufacturing company specializing in **motors, controllers, powertrains, wiring harnesses, SCADA automation and IoT solutions.** We develop customized solutions for **electric mobility, industrial automation, agriculture, renewable energy, Oil & Gas, aerospace, defence** and other demanding applications.

Our 50,000 sq. ft. manufacturing facility is equipped with modern machinery, engineering laboratories and advanced testing facilities. With strong in-house engineering capabilities, we manage the complete product journey from design and prototyping to testing, validation and production. We work closely with OEMs to provide high-performance, reliable and application-specific solutions.

Our growth is driven by innovation, engineering excellence, quality, integrity and long-term partnerships with our customers, suppliers and employees.

Vision

To become the preferred global technology partner for high-efficiency motors, controllers and powertrains from agriculture to aviation by delivering innovation, quality, reliability, on-time service and environmentally responsible manufacturing.

Mission

To become the preferred global technology partner for high-efficiency motors, controllers and powertrains from agriculture to aviation by delivering innovation, quality, reliability, on-time service and environmentally responsible manufacturing.

Core Values

Customer Success | Innovation | Excellence | Integrity | People & Partnerships | Environmental Responsibility.

1. Product Overview

The YBX3 series is an optimized design based on YB3 high-voltage flameproof three-phase induction motors. Motor losses are further reduced, and efficiency meets the Grade 2 energy-efficiency requirements of GB 30254-2013. The series features a compact structure, light weight, high efficiency, energy savings, low noise and vibration, high reliability, long service life, and convenient installation and maintenance.

Standard mounting is IMB3 (foot-mounted). Under normal operating conditions, the motor operates in continuous duty (S1), at ambient temperatures from −20°C to +40°C and at altitudes not exceeding 1,000 m.

1. Basic Motor Specifications

Parameter	Specification
Frame size	355–630
Synchronous speed	3000, 1500, 1000, 750, 600 r/min
Rated frequency	50 Hz; 60 Hz available on request (except 2-pole motors)
Rated voltage	6 kV, 10 kV; other voltages on request
Duty	S1 continuous duty
Power range	160–3150 kW
Explosion protection	Ex d I Mb; Ex d IIA T4 Gb; Ex d IIB T4 Gb; optional Ex d IIC T4 Gb (not available with sleeve bearings)
Cooling	IC411
Ingress protection	IP54; optional IP55, IP56, IP65
Environmental options	TH, THW, TA, TAW, T, TW; indoor F1/F2; outdoor W/WF1/WF2; tropical TH/WTH

Applications: Fans, pumps, compressors, crushers, cutting machines, conveyors and other general-purpose mechanical equipment in petroleum, chemical, coal, power generation, metallurgy, transportation, textiles, pharmaceuticals and food/grain processing industries.

2. Structural Features

Compact Installation and Higher Performance

By reducing installation space and motor dimensions, the YBX3 series achieves compact construction while maintaining higher performance and reliability. The optimized high-voltage flameproof three-phase induction motor meets these requirements with dependable operation.

Improved Reliability and Longer Service Life

The frame and end shields are manufactured from high-strength cast iron using mature production processes. Aerodynamic optimization improves cooling airflow and heat dissipation.

The YBX3 flameproof motor frame uses horizontal, fully distributed cooling fins with a narrowed central section. Fin spacing and concentric-ring arrangements are optimized, with four internal axial ventilation passages. Axially aligned fins at the winding ends increase heat-dissipation area. Corrosion-resistant protective treatments and surface coatings improve durability under climatic exposure. Dual-circuit cooling supports high power density and uniform internal temperature distribution, extending motor service life.

Optimized Cooling System and Airflow Design

The YBX3 series employs an optimized cooling-fin arrangement. As motors become more compact and thermal loading increases, improved heat dissipation is essential. The cooling system is designed for effective heat removal during operation.

The motor has two independent airflow paths: an internal circuit and an external circuit. Much of the generated heat is transferred through the stator core to the frame and removed by external cooling air. The remaining heat, primarily from rotor losses and stator winding-end losses, is carried to the frame by internal airflow. External air passes along the cooling fins from the non-drive end toward the drive end. The terminal box is positioned away from the main high-velocity cooling airflow to reduce direct air impact.

The inner side of the motor frame, around the stator winding ends, is provided with internal cooling fins that effectively improve heat dissipation. The internal airflow is driven by the fan at the non-drive end through rotor ventilation holes. The air flows rapidly over the inner cooling fins of the frame and end shields, providing additional cooling. Heat exchange between the frame and the outside air further improves the cooling effect. Four air channels direct the flow toward the drive end, resulting in more uniform heat dissipation through the internal and external circuits (see Figures 1 and 2).

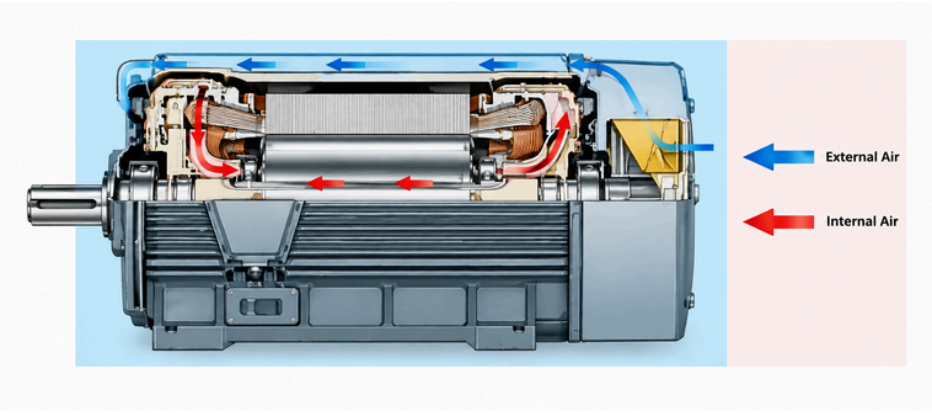


Figure 1. Schematic of the motor dual-circuit cooling system

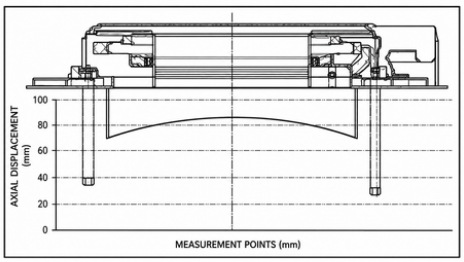


Figure 2. Temperature distribution of motor windings and bearings

2.4 Lower Vibration and Noise

Reduced Vibration

An optimized rotor dynamic design, innovative manufacturing processes and high-precision dynamic balancing ensure low vibration levels. The YBX3 series high-efficiency high-voltage flameproof three-phase induction motors not only meet the vibration requirements of GB 10068, but also achieve levels substantially better than the standard in most applications.

Lower Noise

The optimized airflow design is inseparable from the correct choice of fan type. In IC411-cooled motors, the airflow outside the motor is mainly axial along the spaces between the cooling fins. Therefore, the fan and fan cover are designed with an optimized streamlined profile.

a) External fan

All 2-pole motors use axial-flow external fans, which are suitable for only one direction of rotation. Fan blades comply with GB 3836 and provide adequate resistance to aging and corrosion.
For H355–H400 motors with 4 or more poles, backward-curved centrifugal fans are used and can rotate in either direction. The fans are manufactured from cast aluminum alloy ZL102 or welded steel plate (see Figure 3).

For H450 and larger motors with 4 poles, backward-curved centrifugal fans are used and are suitable for one direction of rotation only. The fans are welded from steel plate. Other multi-pole motors use centrifugal fans capable of bidirectional rotation. The fans are manufactured from cast aluminum alloy ZL102 or welded steel plate (see Figure 3).

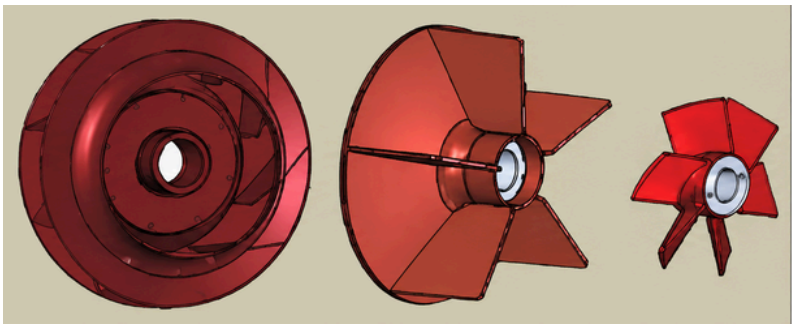


Figure 3. Centrifugal external fans

b) Internal fan
All internal fans are bidirectional centrifugal fans cast from aluminum alloy ZL102 (see Figure 4). This design eliminates the need to dismantle the motor to replace the internal fan when reversing the rotation direction of a conventional 2-pole motor. For 2-pole motors requiring reverse rotation, only the external fan needs to be replaced.

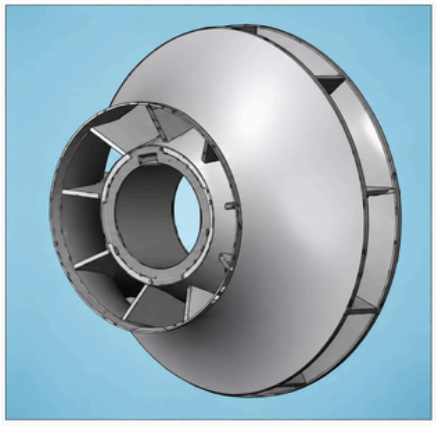


Figure 4. Cast-aluminum internal fan

The fan cover incorporates noise-reduction features and has strong impact resistance. To reduce motor noise, the motor fan cover is fitted with a noise-reducing structure. In addition, the appropriate selection of stator and rotor slot combinations and other low-noise components helps ensure a lower overall motor noise level.

2.5 Frame and End-Shield Structure

The motor frame and end shields are made of high-quality cast iron of grade HT250 or above, using established manufacturing processes. Each motor center height corresponds to a specific frame, with integral cast feet. The optimized frame design provides excellent strength and rigidity. The horizontally or vertically mounted cast-iron frame features full-length cooling ribs and four axial ventilation passages. Axial parallel cooling fins are arranged around the winding end regions inside the frame to increase heat-dissipation area. The terminal box can be mounted on either side of the motor. The drive-end and non-drive-end end shields have the same structure, with internal and external cooling ribs to improve rigidity and cooling performance (Figure 5).

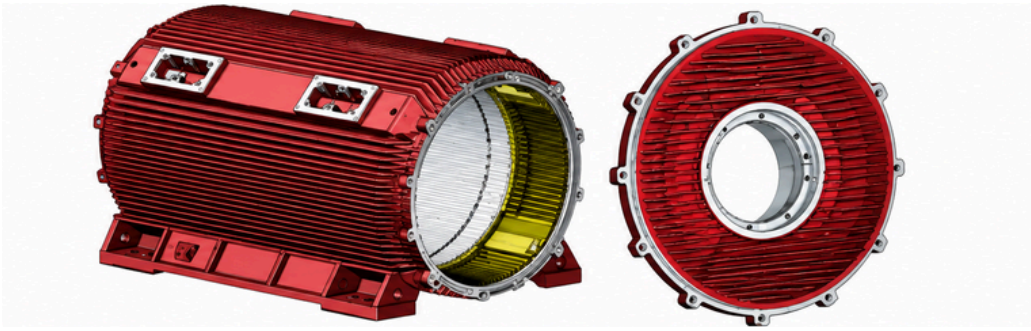


Figure 5. Motor frame and end shield

2.6 Reliable Squirrel-Cage Rotor Structure

The YBX3 series uses a cast-aluminum rotor. Molten aluminum is poured into the rotor core slots using centrifugal casting technology and formed in one operation, integrating the rotor bars and end rings into a single assembly. The cast-aluminum rotor construction and manufacturing process ensure excellent rotor reliability and satisfactory operating characteristics.

2.7 Reliable Bearing Structure

Bearings are available in rolling-bearing and sleeve-bearing configurations. Rolling bearings are generally used; sleeve bearings are selected when operating conditions, such as speed, load capacity or duty requirements, exceed the applicable limits. Standard-design H500-2 (6 kV, 10 kV), H560-2 (6 kV, 10 kV) and H630-2 (6 kV, 10 kV) motors use sleeve bearings with forced lubrication; H500-2 may also be configured with self-lubricating sleeve bearings. Other motors use rolling bearings with continuous oil filling and draining arrangements.

Rolling-bearing assemblies include the end shield, bearing housing, inner and outer covers, bearing, oil slinger, retaining ring, continuous oil supply and drainage devices, and bearing-temperature monitoring. Oil-cup lubrication uses lithium-based grease. V-type sealing rings prevent dust and water from entering the bearings, extending service life and lubrication intervals. The calculated bearing service life exceeds 100,000 hours.

The drive-end and non-drive-end rolling-bearing arrangements are illustrated below. The drive end uses one deep-groove ball bearing and one cylindrical roller bearing; the non-drive end uses one cylindrical roller bearing (Figure 6a and 6b).

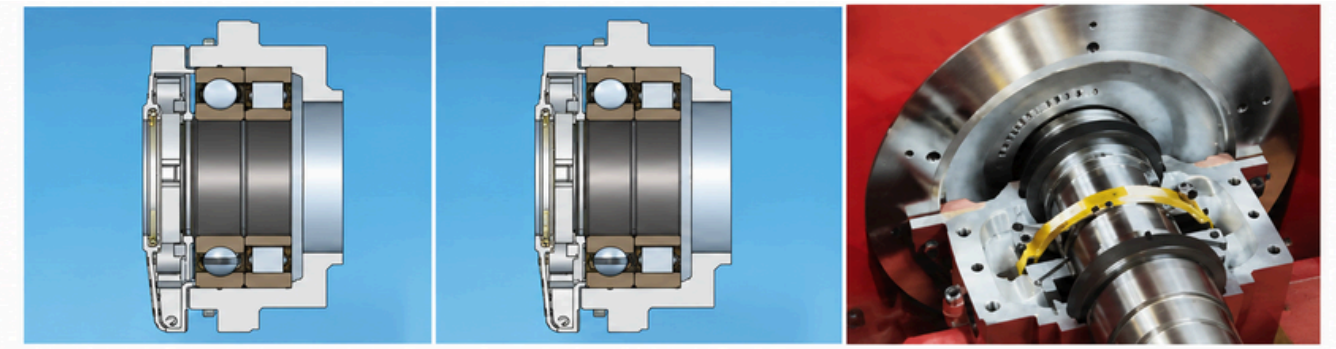


Figure 6. Bearing system: (a) drive-end rolling bearing; (b) non-drive-end rolling bearing; (c) end-shield sleeve bearing.

Sleeve bearings are generally selected according to operating conditions or customer requirements. The sleeve-bearing assembly comprises the bearing cover, end shield, bearing shell, and bearing-temperature measurement device (Figure 6c). Depending on bearing load, self-lubrication by heat radiation and natural convection or forced lubrication can be selected. Self-lubrication may be changed to forced lubrication without modification.

Both drive-end and non-drive-end bearings are normally end-shield-mounted spherical sleeve bearings. The non-shaft-extension-end bearing is insulated. Where the motor shaft extension and coupling are not subjected to axial thrust and axial movement is within ±5 mm, the motor rotor and shaft system must be axially positioned using the driven equipment's limiting device.

2.8 Proven VPI Vacuum Pressure Impregnation — Class 155 (F) Insulation

The insulation system used in our high-voltage motors has been proven over many years of service. By combining this experience with new technologies and modern manufacturing processes, the proportion of mica has been progressively increased, and the insulation system continues to be developed and improved.

Insulation and Terminal Box

The stator winding uses Class 155 (F) insulation. The winding ends are firmly secured and tied. During manufacturing, multiple inter-turn impulse voltage and power-frequency withstand-voltage tests are performed. The VPI (vacuum pressure impregnation) solvent-free process provides high electrical and mechanical strength, corrosion resistance and moisture resistance. Special anti-corona treatment of the winding core allows safe operation at higher altitudes.

Low-loss, high-efficiency silicon steel laminations are used for the punched core to reduce iron losses and improve efficiency and operating reliability. The optimized slot geometry and suitable magnetic slot wedges can reduce iron loss by an average of 20%, increase efficiency by over 0.3%, improve power factor and reduce stator temperature rise.

Reliable and Versatile Terminal Box Arrangements

The motor external leads are reliably connected through terminals inside the terminal box (Figure 7). The main power terminal box is a flameproof cast-iron unit with an attractive, lightweight and standardized design. Protection up to IP55 is available. Its large interior provides adequate electrical clearance and creepage distance in accordance with relevant national standards. The inside is coated with anti-rust primer and corrosion-resistant paint.

Electroplated terminals connect to the winding lead cables. Compression-type fittings secure the outgoing cables. Cable entries include sealing and anti-pullout devices. Available concentric sealing-ring diameters include Ø64, Ø57, Ø51, Ø46 and Ø40 mm. Independent earthing terminals are provided inside and outside the box; the earthing stud diameter is M12.

For 6 kV motors, the terminal box can be installed in 30° increments, offering several permissible installation positions (Figure 8). When installation space is restricted or special cable routing is required, this flexibility is useful. The symmetrical lead arrangement permits repositioning the box from the right to the left side, viewed from the shaft end. An additional terminal box can be provided for star-point leads. Stator and bearing temperature measurement and anti-condensation heater wiring can be housed in a separate auxiliary terminal box. Cable outlet seals are available with concentric diameters Ø20, Ø16, Ø12 and Ø8 mm.

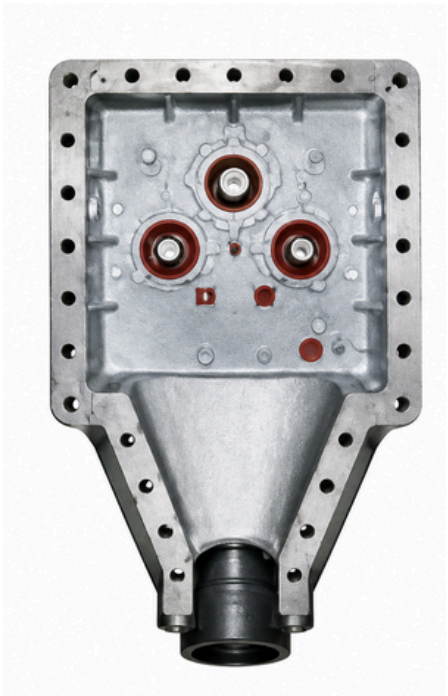


Figure 7. Cast-iron terminal box

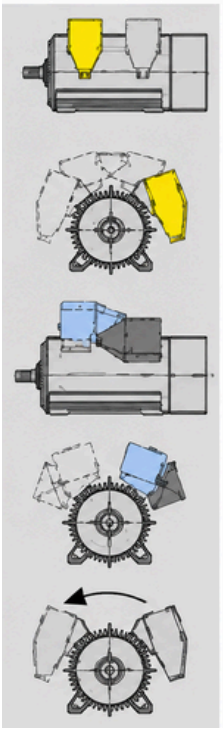


Figure 8. Multiple terminal-box mounting arrangements

Lifting, Safety and Corrosion Protection

2.9 Lifting and Safety

For safe lifting and installation, two lifting lugs must be installed diagonally on the motor frame (Figure 9). The lifting rings can rotate through 360°, and the shackles can turn through 180°. They automatically align with the direction of lifting force.

The frame feet include threaded holes for jack screws to adjust the installation height. Large-area grounding terminals are provided on both sides of the frame base and inside and outside the terminal box (Figure 10).



Figure 9. Lifting lug position



Figure 10. M12 grounding terminal

2.10 Corrosion Protection and Monitoring

Anti-Corrosion Treatment

Surface treatment is suitable for severe environmental conditions, including:
a) Humidity and alternating wet/dry conditions.
b) Low or high temperatures.
c) Ultraviolet radiation and radioactive radiation.
d) Corrosive media, including salt spray, corrosive gases and vapors.

The corrosion-protection system helps prevent these hazards. Depending on the operating environment, standard or special coating systems may be selected. Both systems consist of primer, intermediate coating and topcoat and are suitable for indoor or outdoor installation.

These coatings are dense, wear-resistant, resistant to ageing, flexible yet tough, and pressure-resistant. To obtain good adhesion and long service life, the surface must be carefully prepared before coating. The casting surface is thoroughly cleaned, followed by application of primer.

A uniform and reliable anti-corrosion coating, including at corners and edges, can only be achieved using specialized processing methods.

The primer is applied using this method, and the first finishing coat is completed in the same way. The intermediate coat and second finishing coat are applied by manual high-pressure airless spraying.

Monitoring and Protection

The service life and quality of a motor are influenced by factors that may not be fully addressed during design. Temperature-sensing elements are installed for continuous monitoring. Pt100 resistance temperature detectors (RTDs) are used for stator winding and bearing temperature measurement. An anti-condensation space-heating system is fitted; its supply is normally 220 V AC, with 127 V available for mining motors

Quality Assurance

- Components are manufactured in accordance with quality plans and quality-control requirements.
- In-process inspections are performed at every stage of manufacture.
- Outsourced and purchased components undergo appropriate inspection and testing.
- Final product inspection and performance testing are carried out.
- After-sales service is provided in accordance with the company's Quality Assurance Statement.

3. TECHNICAL DATA FOR PRODUCT SELECTION AND INSTALLATION DIMENSIONS

Motor Model Designation

Example: YBX3 – 500 2 – 4
YBX3: Y = three-phase asynchronous motor; B = flameproof; X = high efficiency; 3 = third-generation design.
500: shaft center height of 500 mm. 2: two iron cores. 4: four poles.

The primer is applied using this method, and the first finishing coat is completed in the same way. The intermediate coat and second finishing coat are applied by manual high-pressure airless spraying.

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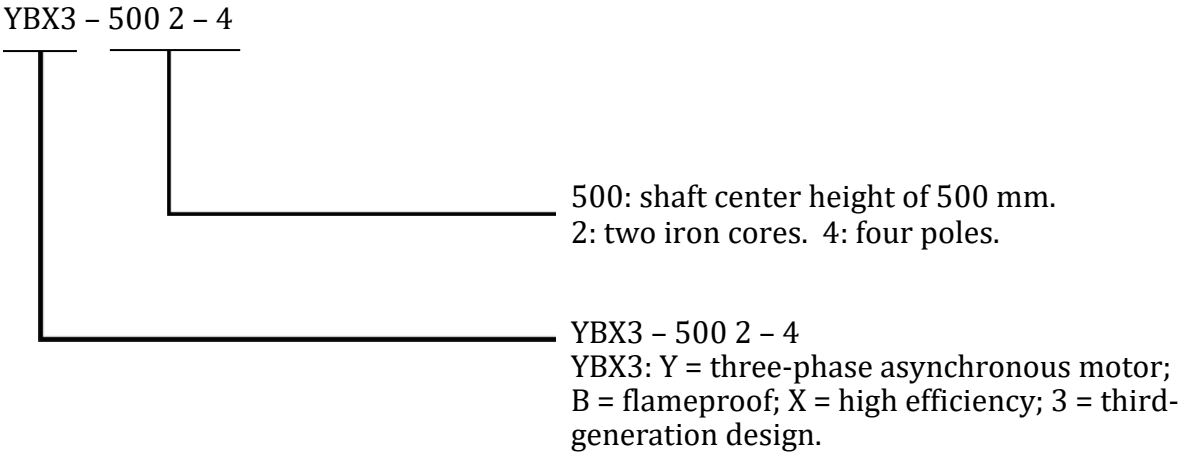
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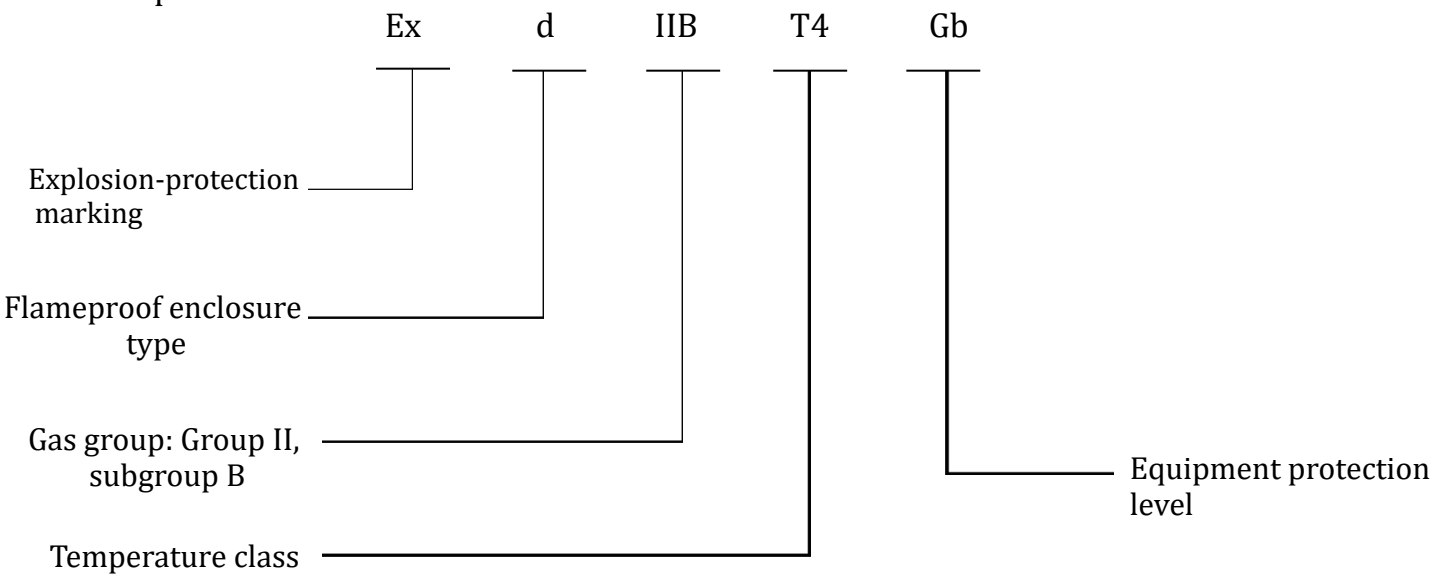
Motor Model Designation

Example:



Meaning of the Explosion-Protection Marking

Example 2:



This series may also be produced in YB3-TH humid-tropical and YB3-W outdoor variants, as well as F, F1 and F2 corrosion-protection variants.

Meaning of Protective-Type Designations

W — Outdoor, light corrosion resistance
WF1 — Outdoor, medium corrosion resistance
WF2 — Outdoor, strong corrosion resistance

F — Indoor, light corrosion resistance
F1 — Indoor, medium corrosion resistance
F2 — Indoor, strong corrosion resistance
TH — Humid tropical
TA — Dry tropical
WTH — Outdoor humid tropical
WTA — Outdoor dry tropical

The wide-range series shows the power range in a stepped format.

Available rated voltages: 6 kV and 10 kV; rated frequencies: 50 Hz and 60 Hz; pole counts: 2–10. Explosion-protection markings Ex d I Mb, Ex d IIA T4 Gb and Ex d IIB T4 Gb may be selected as standard. Ex d IIC T4 Gb motors are available subject to consultation with the manufacturer.

4. Basic Selection

Factors to Consider When Selecting a Three-Phase Induction Motor

Operating environment: ambient temperature, altitude, etc.
Mechanical characteristics and moment of inertia of the driven equipment.
Starting method, starting frequency, and voltage drop during starting.
Temperature-rise assessment.
Direction of motor rotation.
Mechanical coupling method and requirements between motor and driven load.
Terminal box position.
Duty cycle and other special requirements.

4Standard Ordering

Orders without special requirements are treated as standard orders. The following motor parameters must be specified when placing an order (example):

Parameter	Specification
Model	YBX3 560-2
Rated Output	1600 kW
Rated Voltage	6 kV
Rated Frequency	50 Hz
Number of Poles	2
Mounting Arrangement	B3
Duty Cycle	S1 (Continuous)
Cooling Method	IC411
Degree of Protection	IP54
Thermal Class	155 (F)
Explosion Protection	Ex d IIB T4 Gb
Rotation Direction	Clockwise

Operating Environment	Indoor
Cable Entry	Rubber-sheathed cable
Altitude	≤ 1000 m
- 20 C to 40 C	- 20 C to 40 C

Ordering and Technical Data

4.3 Orders with Additional Requirements

If any of the following conditions or requirements apply, they must be clearly stated when ordering:

- Rated voltage other than 6 kV or 10 kV, such as 3.3 kV or 6.3 kV.
- Rated frequency other than 50 Hz, such as 60 Hz.
- Rated output other than the standard value in the data tables, or a different relationship between output and mounting dimensions.
- Altitude above 1000 m.
- Ambient temperature above +40°C or minimum temperature below −20°C.
- Low-noise requirement below the limits of GB 10069.3.
- Non-standard mounting dimensions or frame-to-mounting relationships.
- Special environments, including humid tropical, outdoor, corrosive, salt fog, mould, dust, sand, etc. Identify corrosive substances so suitable protective measures can be designed.
- Special performance requirements, such as high locked-rotor torque, high slip, high starting inertia, or low locked-rotor current.
- Requirement for bidirectional rotation.
- Other special requirements, including winding or bearing temperature monitoring, differential protection, special bearings, special packaging, or non-standard accessories.

If the motor is driven by a variable-frequency drive, specify this when ordering.
For requirements not covered above, consult the company technical department and agree on technical specifications.

As technology advances and applicable national standards are revised, dimensions and technical parameters in this catalogue may change without separate notice. After ordering, refer to the official outline and installation dimension drawings.

5. Technical Data (Guaranteed Values)

- YBX3 series flameproof high-voltage three-phase induction motor type and performance data (6 kV) — see Table 1.
- YBX3 series flameproof high-voltage three-phase induction motor type and performance data (10 kV) — see Table 2.
- YBX3 series flameproof high-voltage three-phase induction motor technical data (6 kV) — see Table 3.
- YBX3 series flameproof high-voltage three-phase induction motor technical data (10 kV) — see Table 4.

Table 1. YBX3 Series Motor Selection Chart (6 kV)

Frame Size	Model No.	Synchronous Speed (r/min)				
		3000	1500	1000	750	600
		Rated Power (kW)				
355	1	185	185	-	-	-
	2	200	200	-	-	-
	3	220	220	185	-	-
	4	250	250	200	-	-
	5	280	280	220	185	-
	6	315	315	250	200	-
	7	355	355	280	220	-
	8	400	400	315	250	-
	9	(450)	(450)	(355)	(280)	-
400	1	450	450	355	280	185
	2	500	500	400	315	200
	3	560	560	450	355	220
	4	(630)	(630)	500	400	250
	5	-	-	(560)	-	280
450	1	630	630	560	450	315
	2	710	710	630	500	355
	3	800	800	710	560	400
	4	(900)	900	(800)	(630)	450
	5	-	(1000)	-	-	-
500	1	900	1000	800	630	500
	2	1000	1120	900	710	560
	3	1120	1250	1000	800	630
	4	1250	1400	1120	(900)	(710)
	5	1400	(1600)	(1250)	-	-
560	1	1600	1600	1250	900	710
	2	1800	1800	1400	1000	800
	3	2000	2000	1600	1120	900
	4	-	2240	1800	1250	1000
	5	-	-	-	1400	1120
630	1	2240	2500	2000	1600	1250
	2	2500	2800	2240	1800	1400
	3	2800	3150	2500	2000	1600
	4	-	-	-	2240	1800

Table 2. YBX3 Series Motor Selection Chart (10 kV)

Frame Size	Model No.	Synchronous Speed (r/min)				
		3000	1500	1000	750	600
		Rated Power (kW)				
400	1	185	185	-	-	-
	2	200	200	-	-	-
	3	220	220	185	-	-
	4	250	250	200	-	-
	5	280	280	220	-	-
	6	315	315	250	185	-
	7	355	355	280	200	-
	8	400	400	315	220	185
	9	450	450	355	250	200
	10	(500)	500	400	280	220
	11	-	(560)	(450)	315	250
450	1	500	560	450	355	280
	2	560	630	500	400	315
	3	630	710	560	450	355
	4	710	800	630	500	400
	5	(800)	(900)	(710)	-	-
500	1	800	900	710	560	450
	2	900	1000	800	630	500
	3	1000	1120	900	710	560
	4	1120	1250	1000	800	630
	5	-	-	(1120)	-	-
560	1	1250	1400	1120	900	710
	2	1400	1600	1250	1000	800
	3	1600	1800	1400	1120	900
	4	1800	2000	1600	1250	1000
	5	-	-	-	-	-
630	1	2000	2240	1800	1400	1120
	2	2240	2500	2000	1600	1250
	3	2500	2800	2240	1800	1400
	4	-	-	-	2000	1600

Table 3 YBX3 Series Motor Efficiency and Power Factor Standards (6 kV)

Synchronous speed: r/min | Efficiency: η (%) | Power factor: cos φ

Rated Power (kW)	Synchronous Speed (rpm)									
	Efficiency η (%)					Power Factor (cos φ)				
	3000	1500	1000	750	600	3000	1500	1000	750	600
185	94.2	94.5	94.4	94.3	93.7	0.85	0.85	0.81	0.77	0.77
200	94.3	94.6	94.5	94.4	93.9			0.83	0.78	0.78
220	94.4	94.7	94.6	94.5	94.0					
250	94.5	94.8	94.8	94.6	94.1					
280	94.7	94.9	94.9	94.8	94.3				0.79	
315	94.9	95.0	95.1	94.9	94.4	0.87	0.86	0.84		0.81
355	95.1	95.2	95.3	95.0	94.6					
400	95.4	95.3	95.3	95.2	94.9				0.80	
450	95.6	95.5	95.6	95.3	95.0				0.81	
500	95.8	95.6	95.8	95.6	95.1	0.88	0.87			
560	95.9	95.8	95.9	95.7	95.2					
630	96.0	96.0	96.0	95.8	95.4				0.82	0.81
710	96.1	96.2	96.2	95.9	95.5					
800	96.3	96.2	96.2	96.0	95.7	0.89				0.82
900	96.4	96.3	96.3	96.1	95.8					
1000	96.5	96.4	96.4	96.2	95.9			0.85		
1120	96.6	96.5	96.5	96.3	96.0				0.83	
1250	96.7	96.6	96.6	96.3	96.2	0.90	0.88			
1400	96.8	96.7	96.7	96.4	96.3					
1600	96.9	96.8	96.8	96.5	96.3					0.83
1800	97.0	96.9	96.9	96.6	96.4			0.87		
2000	97.1	97.0	97.0	96.7	–	0.91	0.89		0.84	
2240	97.2	97.1	97.0	96.8	–					
2500	97.2	97.2	97.1	–	–					–
2800	97.3	97.2	–	–	–			–	–	
3150	–	97.3	–	–	–	–				

Note: Power factor values shown in merged cells in the original chart have been expanded into individual rows for editing. Verify against the manufacturer's official data before use.

Table 4 YBX3 Series Motor Efficiency and Power Factor Standards (10 kV)

Synchronous speed: r/min | Efficiency: η (%) | Power factor: cos

Rated Power (kW)	Synchronous Speed (rpm)									
	Efficiency η (%)					Power Factor (cos φ)				
	3000	1500	1000	750	600	3000	1500	1000	750	600
185	94.2	94.2	93.9	93.6	93.5	0.84	0.83	0.82	0.77	0.74
200	94.3	94.3	94.0	93.7	93.6		0.84			
220	94.4	94.4	94.1	93.8	93.7				0.78	0.75
250	94.5	94.5	94.2	94.0	93.9	0.85				
280	94.7	94.6	94.4	94.2	94.3					
315	94.9	94.8	94.6	94.6	94.5				0.79	0.76
355	95.2	94.9	94.7	94.7	94.6					
400	95.4	95.0	94.9	94.9	94.7	0.86	0.86	0.83		
450	95.6	95.4	95.0	95.0	94.9				0.80	
500	95.7	95.4	95.4	95.3	95.0					0.78
560	95.8	95.6	95.5	95.4	95.1					
630	95.8	95.8	95.8	95.8	95.3	0.87			0.81	
710	96.0	96.2	95.9	95.9	95.5		0.87			0.80
800	96.1	96.2	96.0	96.0	95.7					
900	96.2	96.3	96.2	96.1	95.9	0.88		0.84		
1000	96.3	96.4	96.3	96.3	95.9					0.81
1120	96.4	96.5	96.5	96.4	96.1				0.82	
1250	96.6	96.6	96.6	96.5	96.1	0.89		0.85		
1400	96.7	96.8	96.8	96.5	96.1					
1600	96.7	96.9	96.9	96.5	96.1					
1800	96.8	97.0	96.9	96.6	-	0.91	0.90	0.86		
2000	96.9	97.1	96.9	96.6	-					
2240	97.1	97.2	96.9	-	-					
2500	97.1	97.2	-	-	-					
2800	-	97.2	-	-	-	-	0.91	-	-	-

Table 3 YBX3 Series High-Voltage High-Efficiency Three-Phase Induction Motor Technical Data (6 kV)

Model	Rated Power	Rated Current	Synchronous Speed	Efficiency η	Power Factor	Locked Rotor Torque (Multiple of Rated Torque)	Breakdown Torque (Multiple of Rated Torque)	Pull-out Torque (Multiple of Rated Torque)	Weight
	kW	A	r/min	%	cos φ				kg
YBX3 3551-2	185	22.2	3000	91.2	0.85	2.0	0.7	7.0	2450
YBX3 3552-2	200	24.0	3000	91.3	0.85	2.0	0.7	7.0	2500
YBX3 3553-2	220	26.4	3000	91.4	0.85	2.0	0.7	7.0	2550
YBX3 3554-2	250	30.0	3000	91.5	0.85	2.0	0.7	7.0	2650

Model	Rated Power	Rated Current	Synchronous Speed	Efficiency η	Power Factor	Starting Torque (Locked Rotor) Multiple of Rated Torque	Starting Current Multiple of Rated Current	Maximum Torque Multiple of Rated Torque	Weight
	kW	A	rpm	%	cos φ				kg
YBX3 3555-2	280	32.7	3000	94.7	0.87	2.0	0.7	7.0	2750
YBX3 3556-2	315	36.7	3000	94.9	0.87	2.0	0.7	7.0	2850
YBX3 3557-2	355	41.3	3000	95.1	0.87	2.0	0.7	7.0	2950
YBX3 3558-2	400	46.4	3000	95.4	0.87	2.0	0.7	7.0	3050
YBX3 4001-2	450	51.5	3000	96.6	0.88	2.0	0.7	7.0	3250
YBX3 4002-2	500	57.1	3000	95.8	0.88	2.0	0.7	7.0	3450
YBX3 4003-2	560	63.9	3000	95.9	0.88	2.0	0.7	7.0	3650
YBX3 4501-2	630	71.8	3000	96.0	0.88	2.0	0.7	7.0	4400
YBX3 4502-2	710	80.8	3000	96.1	0.88	2.0	0.7	7.0	4700
YBX3 4503-2	800	89.8	3000	96.3	0.89	2.0	0.7	7.0	5000
YBX3 5001-2	900	100.9	3000	96.4	0.89	2.0	0.7	7.0	5700
YBX3 5002-2	1000	112.0	3000	96.5	0.89	2.0	0.7	7.0	6000
YBX3 5003-2	1120	125.4	3000	96.6	0.89	2.0	0.7	7.0	6300
YBX3 5004-2	1250	138.2	3000	96.7	0.90	2.0	0.7	7.0	6600
YBX3 5005-2	1400	154.6	3000	96.8	0.90	2.0	0.7	7.0	6900
YBX3 5601-2	1600	176.5	3000	96.9	0.90	2.0	0.7	7.0	8700
YBX3 5602-2	1800	198.4	3000	97.0	0.90	2.0	0.7	7.0	9100
YBX3 5603-2	2000	217.8	3000	97.1	0.91	2.0	0.7	7.0	9500
YBX3 6301-2	2240	243.7	3000	97.2	0.91	2.0	0.7	7.0	11500
YBX3 6302-2	2500	272.0	3000	97.2	0.91	2.0	0.7	7.0	12100
YBX3 6303-2	2800	304.3	3000	97.3	0.91	2.0	0.7	7.0	12700
YBX3 3551-4	185	22.2	1500	94.5	0.85	2.0	0.8	6.5	2450
YBX3 3552-4	200	23.9	1500	94.6	0.85	2.0	0.8	6.5	2500
YBX3 3553-4	220	26.3	1500	94.7	0.85	2.0	0.8	6.5	2600
YBX3 3555-4	250	29.9	1500	94.8	0.85	2.0	0.8	6.5	2700
YBX3 3556-4	315	37.1	1500	95.0	0.86	2.0	0.8	6.5	2800
YBX3 3557-4	355	41.7	1500	95.2	0.86	2.0	0.8	6.5	3000
YBX3 3558-4	400	47.0	1500	95.3	0.86	2.0	0.8	6.5	3100
YBX3 4001-4	450	52.7	1500	95.5	0.86	2.0	0.8	6.5	3300

YBX3 Series Motor Technical Data (6 kV) — Continued

Motor Model	Rated Power (kW)	Stator Current (A)	Synchronous Speed (r/min)	Efficiency (%)	Power Factor (cosφ)	Maximum Torque Ratio	Locked-Rotor Torque Ratio	Locked-Rotor Current Ratio	Weight (kg)
YBX3 4002-4	500	57.8	1500	95.6	0.87	2.0	0.8	6.5	3500
YBX3 4003-4	560	64.7	1500	95.8	0.87	2.0	0.8	6.5	3700
YBX3 4501-4	630	72.6	1500	96.0	0.87	2.0	0.8	6.5	4300
YBX3 4502-4	710	81.6	1500	96.2	0.87	2.0	0.8	6.5	4550
YBX3 4503-4	800	92.0	1500	96.2	0.87	2.0	0.8	6.5	4800
YBX3 4504-4	900	103.4	1500	96.3	0.87	2.0	0.8	6.5	5050
YBX3 5001-4	1000	114.7	1500	96.4	0.87	2.0	0.8	6.5	6000
YBX3 5002-4	1120	128.4	1500	96.5	0.87	2.0	0.8	6.5	6300
YBX3 5003-4	1250	141.5	1500	96.6	0.88	2.0	0.8	6.5	6600
YBX3 5601-4	1400	158.3	1500	96.7	0.88	2.0	0.8	6.5	6900
YBX3 5602-4	1600	180.7	1500	96.8	0.88	2.0	0.8	6.5	8100
YBX3 5603-4	1800	203.1	1500	96.9	0.88	2.0	0.8	6.5	8600
YBX3 5604-4	2000	222.9	1500	97.0	0.89	2.0	0.8	6.5	9100
YBX3 6301-4	2240	249.4	1500	97.1	0.89	2.0	0.8	6.5	9600
YBX3 6302-4	2500	278.1	1500	97.2	0.89	2.0	0.8	6.5	11800
YBX3 6303-4	2800	311.5	1500	97.2	0.89	2.0	0.8	6.5	12400
YBX3 6304-4	3150	350.0	1500	97.3	0.89	2.0	0.8	6.5	13000
YBX3 3553-6	185	23.3	1000	94.1	0.81	2.0	0.8	6.0	2550
YBX3 3554-6	200	25.1	1000	94.5	0.81	2.0	0.8	6.0	2600
YBX3 3555-6	220	27.0	1000	94.6	0.83	2.0	0.8	6.0	2700
YBX3 3556-6	250	30.6	1000	94.8	0.83	2.0	0.8	6.0	2800
YBX3 3557-6	280	34.2	1000	94.9	0.83	2.0	0.8	6.0	2900
YBX3 3558-6	315	38.4	1000	95.1	0.83	2.0	0.8	6.0	3000
YBX3 4001-6	355	42.7	1000	95.3	0.84	2.0	0.8	6.0	3350
YBX3 4002-6	400	48.1	1000	95.3	0.84	2.0	0.8	6.0	3500
YBX3 4003-6	450	53.9	1000	95.6	0.84	2.0	0.8	6.0	3650
YBX3 4004-6	500	59.8	1000	95.8	0.84	2.0	0.8	6.0	3600
YBX3 4004-6	500	59.8	1000	95.8	0.84	2.0	0.8	6.0	3800
YBX3 4501-6	560	66.9	1000	95.9	0.84	2.0	0.8	6.0	4150
YBX3 4502-6	630	75.2	1000	96.0	0.84	2.0	0.8	6.0	4500

YBX3 Series High-Voltage High Efficiency Three-Phase Induction Motor (6 kV)

Motor Model	Rated Power (kW)	Stator Current (A)	Synchronous Speed (r/min)	Efficiency (%)	Power Factor (cosφ)	Maximum Torque Ratio	Locked-Rotor Torque Ratio	Locked-Rotor Current Ratio	Weight (kg)
YBX3 4503-6	710	84.5	1000	96.2	0.84	2.0	0.8	6.0	4850
YBX3 5001-6	800	95.3	1000	96.2	0.84	2.0	0.8	6.0	5700
YBX3 5002-6	900	105.8	1000	96.3	0.85	2.0	0.8	6.0	6050
YBX3 5003-6	1000	117.4	1000	96.4	0.85	2.0	0.8	6.0	6400
YBX3 5004-6	1120	131.4	1000	96.5	0.85	2.0	0.8	6.0	6750
YBX3 5601-6	1250	146.5	1000	96.6	0.85	2.0	0.8	6.0	8300
YBX3 5602-6	1400	160.1	1000	96.7	0.87	2.0	0.8	6.0	8700
YBX3 5603-6	1600	182.8	1000	96.8	0.87	2.0	0.8	6.0	9100
YBX3 5604-6	1800	205.5	1000	96.9	0.87	2.0	0.8	6.0	9500
YBX3 6301-6	2000	228.1	1000	97.0	0.87	2.0	0.8	6.0	11700
YBX3 6302-6	2240	255.4	1000	97.0	0.87	2.0	0.8	6.0	12300
YBX3 6303-6	2500	284.8	1000	97.1	0.87	2.0	0.8	6.0	13000
YBX3 3555-8	185	24.6	750	94.3	0.77	2.0	0.8	5.5	2600
YBX3 3556-8	200	26.1	750	94.4	0.78	2.0	0.8	5.5	2700
YBX3 3557-8	220	28.7	750	94.5	0.78	2.0	0.8	5.5	2800
YBX3 3558-8	250	32.2	750	94.6	0.79	2.0	0.8	5.5	2900
YBX3 4001-8	280	36.0	750	94.8	0.79	2.0	0.8	5.5	3350
YBX3 4002-8	315	40.4	750	94.9	0.79	2.0	0.8	5.5	3500
YBX3 4003-8	355	45.5	750	95.0	0.79	2.0	0.8	5.5	3650
YBX3 4004-8	400	50.5	750	95.2	0.80	2.0	0.8	5.5	3800
YBX3 4501-8	450	56.8	750	95.3	0.80	2.0	0.8	5.5	4200
YBX3 4502-8	500	62.1	750	95.6	0.81	2.0	0.8	5.5	4500
YBX3 4503-8	560	69.5	750	95.7	0.81	2.0	0.8	5.5	4800
YBX3 5001-8	630	77.2	750	95.8	0.82	2.0	0.8	5.5	5900
YBX3 5002-8	710	86.9	750	95.9	0.82	2.0	0.8	5.5	6300
YBX3 5003-8	800	97.8	750	96.0	0.82	2.0	0.8	5.5	6700
YBX3 5601-8	900	109.9	750	96.1	0.82	2.0	0.8	5.5	8000
YBX3 5602-8	1000	120.5	750	96.2	0.83	2.0	0.8	5.5	8400

Model	Rated Power	Rated Current	Synchronous Speed	Efficiency η	Power Factor	Starting Torque (Locked Rotor) Multiple of Rated Torque	Starting Current Multiple of Rated Current	Maximum Torque Multiple of Rated Torque	Weight
	kW	A	r/min	%	cos φ				kg
YBX3 5603-8	1120	134.8	750	96.3	0.83	2.0	0.8	5.5	8800
YBX3 5601-8	1250	150.5	750	96.3	0.83	2.0	0.8	5.5	9200
YBX3 5605-8	1400	168.4	750	96.4	0.83	2.0	0.8	5.5	9600
YBX3 6301-8	1600	192.2	750	96.5	0.83	2.0	0.8	5.5	11600
YBX3 6302-8	1800	213.5	750	96.6	0.84	2.0	0.8	5.5	12200
YBX3 6303-8	2000	236.9	750	96.7	0.84	2.0	0.8	5.5	12800
YBX3 6304-8	2240	265.1	750	96.8	0.84	2.0	0.8	5.5	13100
YBX3 4001-10	185	24.7	600	93.7	0.77	2.0	0.8	5.5	3250
YBX3 4002-10	200	26.3	600	93.9	0.78	2.0	0.8	5.5	3350
YBX3 4003-10	220	28.9	600	94.0	0.78	2.0	0.8	5.5	3500
YBX3 4004-10	250	32.8	600	94.1	0.78	2.0	0.8	5.5	3650
YBX3 4005-10	280	36.6	600	94.3	0.78	2.0	0.8	5.5	3800
YBX3 4501-10	315	40.6	600	94.4	0.79	2.0	0.8	5.5	4200
YBX3 4502-10	355	45.7	600	94.6	0.79	2.0	0.8	5.5	4300
YBX3 4503-10	400	51.3	600	94.9	0.79	2.0	0.8	5.5	4500
YBX3 4504-10	450	57.7	600	95.0	0.79	2.0	0.8	5.5	4800
YBX3 5001-10	500	62.5	600	95.1	0.81	2.0	0.8	5.5	5900
YBX3 5002-10	560	69.9	600	95.2	0.81	2.0	0.8	5.5	6300
YBX3 5003-10	630	78.5	600	95.4	0.81	2.0	0.8	5.5	6700
YBX3 5601-10	710	88.3	600	95.5	0.81	2.0	0.8	5.5	8000
YBX3 5602-10	800	98.1	600	95.7	0.82	2.0	0.8	5.5	8400
YBX3 5603-10	900	110.2	600	95.8	0.82	2.0	0.8	5.5	8800
YBX3 5604-10	1000	122.4	600	95.9	0.82	2.0	0.8	5.5	9200
YBX3 5605-10	1120	136.9	600	96.0	0.82	2.0	0.8	5.5	9600
YBX3 6301-10	1250	152.5	600	96.2	0.82	2.0	0.8	5.5	11600
YBX3 6302-10	1400	168.5	600	96.3	0.83	2.0	0.8	5.5	12200
YBX3 6303-10	1600	192.6	600	96.3	0.83	2.0	0.8	5.5	12800
YBX3 6304-10	1800	216.5	600	96.4	0.83	2.0	0.8	5.5	13100

Table 4. YBX3 Series High-Voltage High-Efficiency Three-Phase Induction Motor Technical Data (10 kV)

Model	Rated Power	Rated Current	Synchronous Speed	Efficiency η	Power Factor	Starting Torque (Locked Rotor) Multiple of Rated Torque	Starting Current Multiple of Rated Current	Maximum Torque Multiple of Rated Torque	Weight
	kW	A	r/min	%	cos φ				kg
YBX3 4001-2	185	13.5	3000	94.2	0.84	2.0	0.7	7.0	2950
YBX3 4002-2	200	14.4	3000	94.3	0.85	2.0	0.7	7.0	3080
YBX3 4003-2	220	15.8	3000	94.4	0.85	2.0	0.7	7.0	3180
YBX3 4004-2	250	18.0	3000	94.5	0.85	2.0	0.7	7.0	3290
YBX3 4005-2	280	20.1	3000	94.7	0.85	2.0	0.7	7.0	3390
YBX3 4006-2	315	22.5	3000	94.9	0.85	2.0	0.7	7.0	3480
YBX3 4007-2	355	25.0	3000	95.2	0.86	2.0	0.7	7.0	3590
YBX3 4008-2	400	28.1	3000	95.4	0.86	2.0	0.7	7.0	3670
YBX3 4009-2	450	31.6	3000	95.6	0.86	2.0	0.7	7.0	3770
YBX3 4501-2	500	35.1	3000	95.7	0.86	2.0	0.7	7.0	4350
YBX3 4502-2	560	38.8	3000	95.8	0.87	2.0	0.7	7.0	4500
YBX3 4503-2	630	43.6	3000	95.8	0.87	2.0	0.7	7.0	4850
YBX3 4504-2	710	49.1	3000	96.0	0.87	2.0	0.7	7.0	5100
YBX3 5001-2	800	54.6	3000	96.1	0.88	2.0	0.7	7.0	5860
YBX3 5002-2	900	61.4	3000	96.2	0.88	2.0	0.7	7.0	6200
YBX3 5003-2	1000	67.4	3000	96.3	0.89	2.0	0.7	7.0	6560
YBX3 5004-2	1120	75.4	3000	96.4	0.89	2.0	0.7	7.0	6900
YBX3 5601-2	1250	83.9	3000	96.6	0.89	2.0	0.7	7.0	8400
YBX3 5602-2	1400	93.9	3000	96.7	0.89	2.0	0.7	7.0	8800
YBX3 5603-2	1600	105.0	3000	96.7	0.91	2.0	0.7	7.0	9200
YBX3 5604-2	1800	118.0	3000	96.8	0.91	2.0	0.7	7.0	9600
YBX3 6301-2	2000	131.0	3000	96.9	0.91	2.0	0.7	7.0	11400
YBX3 6302-2	2240	146.4	3000	97.1	0.91	2.0	0.7	7.0	12000
YBX3 6303-2	2500	163.4	3000	97.1	0.91	2.0	0.7	7.0	12600
YBX3 4001-4	185	13.7	1500	94.2	0.83	2.0	0.8	6.5	2960
YBX3 4002-4	200	14.6	1500	94.3	0.84	2.0	0.8	6.5	3120
YBX3 4003-4	220	16.0	1500	94.4	0.84	2.0	0.8	6.5	3200
YBX3 4004-4	250	18.2	1500	94.5	0.84	2.0	0.8	6.5	3280
YBX3 4005-4	280	20.3	1500	94.6	0.84	2.0	0.8	6.5	3360
YBX3 4006-4	315	22.8	1500	94.8	0.84	2.0	0.8	6.5	3440

Motor Model	Rated Power (kW)	Stator Current (A)	Synchronous Speed (r/min)	Efficiency (%)	Power Factor (cosφ)	Maximum Torque Ratio	Locked-Rotor Torque Ratio	Locked-Rotor Current Ratio	Weight (kg)
YBX3 4007-4	355	25.4	1500	94.9	0.85	2.0	0.8	6.5	3520
YBX3 4008-4	400	28.6	1500	95.0	0.85	2.0	0.8	6.5	3600
YBX3 4009-4	450	32.0	1500	95.4	0.85	2.0	0.8	6.5	3700
YBX3 40010-4	500	35.6	1500	95.4	0.85	2.0	0.8	6.5	3800
YBX3 4501-4	560	39.3	1500	95.6	0.86	2.0	0.8	6.5	4150
YBX3 4502-4	630	44.1	1500	95.8	0.86	2.0	0.8	6.5	4650
YBX3 4503-4	710	49.5	1500	96.2	0.86	2.0	0.8	6.5	4850
YBX3 4504-4	800	56.2	1500	96.2	0.87	2.0	0.8	6.5	5050
YBX3 5001-4	900	62.0	1500	96.3	0.87	2.0	0.8	6.5	6000
YBX3 5002-4	1000	68.1	1500	96.4	0.88	2.0	0.8	6.5	6300
YBX3 5003-4	1120	76.1	1500	96.5	0.88	2.0	0.8	6.5	6600
YBX3 5004-4	1250	84.9	1500	96.6	0.88	2.0	0.8	6.5	6900
YBX3 5601-4	1400	94.9	1500	96.8	0.88	2.0	0.8	6.5	8100
YBX3 5602-4	1600	105.9	1500	96.9	0.90	2.0	0.8	6.5	8600
YBX3 5603-4	1800	119.0	1500	97.0	0.90	2.0	0.8	6.5	9100
YBX3 5604-4	2000	132.1	1500	97.1	0.90	2.0	0.8	6.5	9600
YBX3 6301-4	2240	147.8	1500	97.2	0.90	2.0	0.8	6.5	11900
YBX3 6302-4	2500	165.0	1500	97.2	0.90	2.0	0.8	6.5	12400
YBX3 6303-4	2800	182.8	1500	97.2	0.91	2.0	0.8	6.5	13100
YBX3 4003-6	185	13.9	1000	93.9	0.82	2.0	0.8	6.0	3150
YBX3 4004-6	200	15.0	1000	94.0	0.82	2.0	0.8	6.0	3250
YBX3 4005-6	220	16.5	1000	94.1	0.82	2.0	0.8	6.0	3350
YBX3 4006-6	250	18.7	1000	94.2	0.82	2.0	0.8	6.0	3450
YBX3 4007-6	280	20.6	1000	94.4	0.83	2.0	0.8	6.0	3550
YBX3 4008-6	315	23.2	1000	94.6	0.83	2.0	0.8	6.0	3650
YBX3 4009-6	355	26.1	1000	94.7	0.83	2.0	0.8	6.0	3750
YBX3 40010-6	400	29.3	1000	94.8	0.83	2.0	0.8	6.0	3850
YBX3 4501-6	450	33.0	1000	95.0	0.83	2.0	0.8	6.0	4150
YBX3 4502-6	500	36.5	1000	95.4	0.83	2.0	0.8	6.0	4400
YBX3 4503-6	560	40.8	1000	95.6	0.83	2.0	0.8	6.0	4650
YBX3 4504-6	630	45.7	1000	95.8	0.83	2.0	0.8	6.0	4900
YBX3 5001-6	710	51.5	1000	95.9	0.83	2.0	0.8	6.0	5750

Motor Model	Rated Power (kW)	Stator Current (A)	Synchronous Speed (r/min)	Efficiency (%)	Power Factor (cosφ)	Maximum Torque Ratio	Locked-Rotor Torque Ratio	Locked-Rotor Current Ratio	Weight (kg)
YBX3 5002-6	800	57.3	1000	96.0	0.84	2.0	0.8	6.0	6100
YBX3 5003-6	900	64.3	1000	96.2	0.84	2.0	0.8	6.0	6450
YBX3 5004-6	1000	71.4	1000	96.3	0.84	2.0	0.8	6.0	6800
YBX3 5601-6	1120	79.8	1000	96.5	0.84	2.0	0.8	6.0	8150
YBX3 5602-6	1250	87.9	1000	96.6	0.85	2.0	0.8	6.0	8650
YBX3 5603-6	1400	98.2	1000	96.8	0.85	2.0	0.8	6.0	9150
YBX3 5604-6	1600	112.2	1000	96.9	0.85	2.0	0.8	6.0	9650
YBX3 6301-6	1800	126.2	1000	96.9	0.85	2.0	0.8	6.0	11900
YBX3 6302-6	2000	140.2	1000	96.9	0.85	2.0	0.8	6.0	12100
YBX3 6303-6	2240	155.2	1000	96.9	0.86	2.0	0.8	6.0	13000
YBX3 4006-8	185	14.8	750	93.6	0.77	2.0	0.8	5.5	3300
YBX3 4007-8	200	15.8	750	93.7	0.78	2.0	0.8	5.5	3400
YBX3 4008-8	220	17.4	750	93.8	0.78	2.0	0.8	5.5	3500
YBX3 4009-8	250	19.4	750	94.0	0.79	2.0	0.8	5.5	3600
YBX3 40010-8	280	21.7	750	94.2	0.79	2.0	0.8	5.5	3700
YBX3 40011-8	315	24.3	750	94.6	0.79	2.0	0.8	5.5	3800
YBX3 4501-8	355	27.4	750	94.7	0.79	2.0	0.8	5.5	4000
YBX3 4502-8	400	30.4	750	94.9	0.80	2.0	0.8	5.5	4300
YBX3 4503-8	450	34.2	750	95.0	0.80	2.0	0.8	5.5	4600
YBX3 4504-8	500	37.4	750	95.3	0.81	2.0	0.8	5.5	4900
YBX3 5001-8	560	41.8	750	95.4	0.81	2.0	0.8	5.5	5800
YBX3 5002-8	630	46.3	750	95.8	0.82	2.0	0.8	5.5	6100
YBX3 5003-8	710	52.1	750	95.9	0.82	2.0	0.8	5.5	6400
YBX3 5004-8	800	58.7	750	96.0	0.82	2.0	0.8	5.5	6700
YBX3 5601-8	900	65.9	750	96.1	0.82	2.0	0.8	5.5	8150
YBX3 5602-8	1000	72.2	750	96.3	0.83	2.0	0.8	5.5	8650
YBX3 5603-8	1120	80.8	750	96.4	0.83	2.0	0.8	5.5	9150
YBX3 5604-8	1250	90.1	750	96.5	0.83	2.0	0.8	5.5	9650
YBX3 6301-8	1400	100.9	750	96.5	0.83	2.0	0.8	5.5	11200
YBX3 6302-8	1600	115.3	750	96.5	0.83	2.0	0.8	5.5	11900
YBX3 6303-8	1800	128.1	750	96.6	0.84	2.0	0.8	5.5	12500
YBX3 6304-8	2000	142.3	750	96.6	0.84	2.0	0.8	5.5	13200

Motor Model	Rated Power (kW)	Stator Current (A)	Synchronous Speed (r/min)	Efficiency (%)	Power Factor (cosφ)	Maximum Torque Ratio	Locked-Rotor Torque Ratio	Locked-Rotor Current Ratio	Weight (kg)
YBX3 4008-10	185	15.4	600	93.5	0.74	2.0	0.8	5.5	3500
YBX3 4009-10	200	16.7	600	93.6	0.74	2.0	0.8	5.5	3600
YBX3 40010-10	220	18.3	600	93.7	0.74	2.0	0.8	5.5	3700
YBX3 40011-10	250	20.5	600	93.9	0.75	2.0	0.8	5.5	3800
YBX3 4501-10	280	22.9	600	94.3	0.75	2.0	0.8	5.5	4000
YBX3 4502-10	315	25.3	600	94.5	0.76	2.0	0.8	5.5	4300
YBX3 4503-10	355	28.5	600	94.6	0.76	2.0	0.8	5.5	4600
YBX3 4504-10	400	32.1	600	94.7	0.76	2.0	0.8	5.5	4900
YBX3 5001-10	450	35.1	600	94.9	0.78	2.0	0.8	5.5	5800
YBX3 5002-10	500	39.0	600	95.0	0.78	2.0	0.8	5.5	6100
YBX3 5003-10	560	43.6	600	95.1	0.78	2.0	0.8	5.5	6400
YBX3 5004-10	630	48.9	600	95.3	0.78	2.0	0.8	5.5	6700
YBX3 5601-10	710	53.7	600	95.5	0.80	2.0	0.8	5.5	8150
YBX3 5602-10	800	60.3	600	95.7	0.80	2.0	0.8	5.5	8650
YBX3 5603-10	900	67.7	600	95.9	0.80	2.0	0.8	5.5	9150
YBX3 5604-10	1000	75.3	600	95.9	0.80	2.0	0.8	5.5	9650
YBX3 6301-10	1120	84.1	600	96.1	0.80	2.0	0.8	5.5	11200
YBX3 6302-10	1250	92.7	600	96.1	0.81	2.0	0.8	5.5	11900
YBX3 6303-10	1400	103.8	600	96.1	0.81	2.0	0.8	5.5	12500
YBX3 6304-10	1600	118.7	600	96.1	0.81	2.0	0.8	5.5	13200

6. Overall Dimensions and Mounting Dimensions

6.1 Motor overall and mounting dimensions

The overall and mounting dimensions of the motor shall comply with Figure 12 and Table 5.

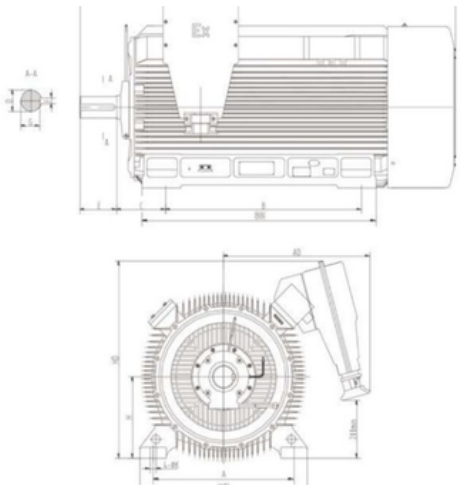


Figure 12. Overall and mounting dimensions of YBX3 series high-voltage flameproof three-phase induction motors (6 kV, 10 kV).

Table 5. YBX3 Series High-Voltage Flameproof Three-Phase Induction Motor Mounting and Overall Dimension Tolerances (6 kV, 10 kV)

Frame Size	Poles	Mounting Dimensions																Overall Dimensions (Max.)				
		A	B	C		D		E		F		G		H		K		AB	AD	BB	HD	L
		Basic Dimension	Basic Dimension	Basic Dimension	Limit Deviation	Basic Dimension	Limit Deviation	Basic Dimension	Limit Deviation	Basic Dimension	Limit Deviation	Basic Dimension	Limit Deviation	Basic Dimension	Limit Deviation	Basic Dimension	Limit Deviation					
355	2	670	1000	254	±1.0	75	+0.035 +0.013	140	±0.5	20	0 -0.052	67.5	0 -0.20	355	- -1.0	28	+0.62 0	840	810	1200	970	2250
	4					100		210	±0.57	28		90										2370
	6					85		170	±0.5	22		76										2370
400	2	750	1120	280	±1.0	85	+0.035 +0.013	170	±0.5	22	0 -0.052	76	0 -0.20	400	0 -1.0	35	+0.62 0	900	980	1340	1110	2370
	4					100		210	±0.57	28		100										2500
	6–10					110		210	±0.57			100										2500
450	2	850	1250	280	±1.0	95	+0.040 +0.015	170	±0.5	25	0 -0.052	86	0 -0.20	450	-	35	-	1030	990	1490	1200	2600
	4					120		210	±0.57	32		109										2600
	6–10					130		250	±0.57	32		119										2680
500	2	950	1330	315	±1.0	110	+0.035 +0.013	210	±0.57	28	0 -0.052	100	0 -0.30	500	0 -1.0	42	+0.62 0	1150	1030	1600	1290	2890
	4					130		250	±0.57	32		119										2810
	6–10					140		250		36		128										2890
560	2	1060	1400	355	±1.0	130	+0.040 +0.015	250	±0.57	32	0 -0.052	119	0 -0.20	560	-	-	-	1270	1050	1800	1390	3250
	4					150		250	±0.57	36		138										3080
	6–10					160		300		40		147										3110
630	2	1180	1600	530	±1.0	140		250	±0.5	36		128	0 -0.30	630	-	48	-	1300	1130	1960	1420	3450
	4					170		300		40		157										3325
	6–10					180		300		45		165										3370

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